# Chapter Exercises

Federico

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### Exercise 1-1

> *Show that $a + b$ can be less than $\min(a, b)$.*

Let $a = -1$ and $b = -1$.

Then $a + b = -2$ and $\min(a, b) = -1$.

Since $-2 < -1$, the claim that $a + b$ can be less than $\min(a, b)$ holds.

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### Exercise 1-2

> *Show that $a \times b$ can be less than $\min(a, b)$.*

Let $a = -1$ and $b = 2$.

Then $a \times b = -2$ and $\min(a, b) = -1$.

Since $-2 < -1$, the claim that $a \times b$ can be less than $\min(a, b)$ holds.

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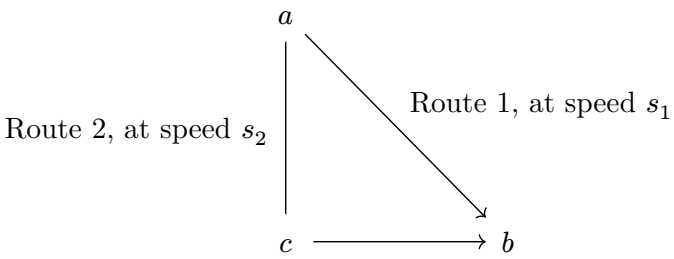
### Exercise 1-3

> *Design/draw a road network with two points $a$ and $b$ such that the fastest route between $a$ and $b$ is not the shortest route.*

We can realize such a network by using speed 'limits'– which should instead be treated as fixed speed specifications.




1. **Route 1 (direct):**
   - By the Pythagorean thourem, its length is $d_1 = \sqrt{2}l$.
   - Let its speed limit (and minimum speed) be $s_1 = v$.
   - The time to traverse it is $t_1 = \frac{d_1}{s_1} = \frac{\sqrt{2}}{v}l$

2. **Route 2 (through $c$):**
   - Its lenght is the sums of the lengths of paths $a$ to $c$ and $c$ to $b$, whicth totals $d_2 = 2l$.
   - Let its speed limit (and minimum speed) be $s_2 = 2v$.
   - The time to traverse it is $t_2 = \frac{d_2}{s_2} = 2\frac{l}{2}v = \frac{l}{v}$

**Conclusion:**
While $d_1 < d_2$, it was shown that $t_1 > t_2$.

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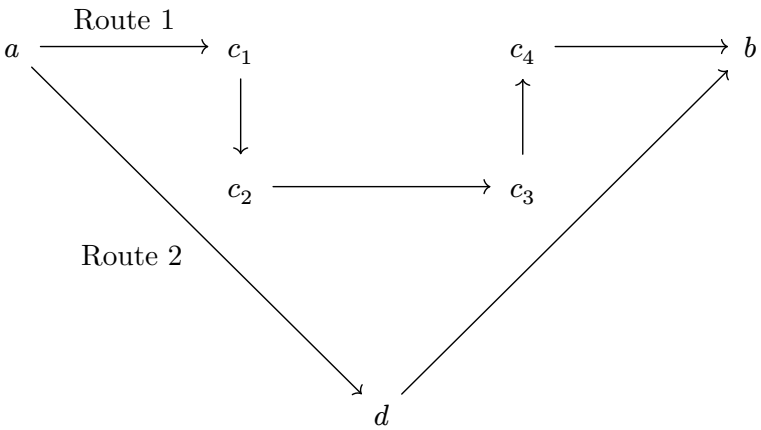
## Exercise 1-4

> *Design/draw a road network with two points a and b such that the shortest route between a and b is not the route with the fewest turns.*

We can go from a to b by taking small deviations, which add turns but amount to little additional distance. This will yield a short path with a high number of turns. It is easy to create a second path with a small number of turns that deviates more from the straight-line path, yielding the desired network.




1. **Route 1:**
   - It's easy to see that the total route length is $d_1 = 2l + l + 2l + l + 2l = 8l$.
   - Number of turns $b_1 = 4$.
2. **Route 2:**
   - By the Pythagorean thourem, its length is $d_2 = 2 \cdot \sqrt{2} \cdot 3l \approx 8.485l$.
   - Number of turns $b_2 = 1$.

**Conclusion:**
While $d_1 < d_2$, it was shown that $b_1 > b_2$.

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